MSX and SPECTRAVIDEO

COMPUTER FORUM

Volume 2 No. 4

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<SCORE>

MSX/SVI : 20, 105

"CHALLENGE STAGE"

IBM C-64 AMSTRAD OLIVETTI ATARI

IBM AMSTRAD OLIVETTI ATARI

OLIVETTI



MEN

200 100 0

POWER

AUSTRALIA'S ONLY MSX & SPECTRAVIDEO COMPUTER MAGAZINE!!!

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COMPUTER FORUM is published by the following people-

MANAGING EDITOR - Ross Hamilton

ASSISTANT EDITOR - Max Cowin

SECRETARY - Tony De-iudicibus

PRODUCTION DESIGNER - Robert Strother

HACKERS COLUMN - Robert Brinkworth

ADVENTURE COLUMN - Alan Crawford

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EDITORIAL

Well here we are again, and it's about time you say!, well I agree with you, I know I stated that we were finally getting the hang of this magazine but I'm afraid getting the hang of it is only a part of what we need, we need help from you in the form of input it makes it very hard to make a magazine if you have no articles, We also require **LISTINGS** we just put our last one's in this issue. So as you can see things are not looking to good in the way of requirements, but I am sure that out there somewhere are some people just bursting to send in some listings & articles.
As I also stated in our last magazine the profits from doing this magazine are hopeless so please don't expect much in the form of payment, we will pay around $5.00 - $10.00 for both listings and articles depending on there merits.

WE WANT YOUR OPINION

We want to hear from you in regards to the magazine, we want you to write to us with your likes or dislikes, what you would like to see in the magazine as well as what you would not like to see.
If you do not wish to put your name on the letter we don't mind but please be honest, we are trying to make a magazine to satisfy everyone, which in itself is an impossible task.

F R O N T C O V E R W I N N E R

This issues front cover winner is -

----- TIM JESSEL of Victoria ------

We think Tim's cover idea was very good and original, so please keep those covers coming.

One last thing if anyone of our readers decide to come to Perth give us a ring and maybe we can get together and have a bit of a chat.

HISTORY OF THE HOME COMPUTER.
SVI & MSX

THIS ARTICLE WAS SENT IN BY ROSS BRISBANE.

It has now been two years since I parted with money, starting me on my computer journey. Lets see, I remember the year 1985 around mid March I purchased the SVI 328 costing $599.00 for computer and $99.00 for the data cassette. THINGS WERE VERY DIFFERENT THEN !!! NOTHING STAYS THE SAME HOWEVER !!! Spectravideo had the market open then. They didn't move fast enough to stop the onslaught of the Commodore 64 unfortunately. In fact they did sweet nothing. Two months later MSX was presented to some 20 companies by Kaye Nishi in Japan. SVI was the catalyst for this standard. Now SVI at this stage deemed the 318 & 328 MSX compatible. (It was felt if there were any inconsistency that this could be rectified by a simple MSX adapter) unfortunately this was not the case... SVI was now faced with a problem. The hardware had been developed already to a stage of disk drives etc...Two directions in which to choose.

Software and the differing standard of MSX hung over the SVI 328 & 318 machines like a long cold icey shadow. We Australians were kept in the dark. I began to build my system into disk drives (C/PM, 80 column etc) while being totally ignorant of SVI's early machines being incompatible with MSX. Yes!! I brought the SVI because of MSX. I knew about the system in March 1985.

SVI attempted to develop an MSX adapter which was to say the least interesting. I had the adapter for review. Its rating for bringing MSX compatibility to SVI 328/318 rated 3/10. In other words a 30% improvement only. WHY? You may ask! The MSX are hard wired very differently. One very different feature was the keyboard layout. This was designed to take into account the multitude of dialects around the world such as Chinese, Russian , Greek , Arabic , Japanese (of course) etc.

MSX computers are popular in the following countries.
Russia , United Kingdom , Holland , Japan , Spain , Italy , Sweden , Germany and a host of other Eastern Block countries. You can't bring SVI 328/318 computers anywhere near the compatibility of an MSX computer at the MACHINE CODE level.

I became upset by all these events. However my interest in this new standard was very intense and as a result was given a SVI 728 MSX computer free of charge to evaluate and give feed-back on.

Cont..

HISTORY OF THE HOME COMPUTER.
SVI & MSX

SVI sent me a letter stating that MSX had become there new direction for devolopment pulling the plug on 328/318 machines. I still feel that they should have continued strong backup support for their earlier machines and there still was a lot of technology life left in the machines . I'm still very much frustrated and disappointed by their failure to us the users of these machines. NO they are not exotic hybrids. They still make sense even today. The failure of this machine to take the market from Commodore is their own fault. Let's not blame the Clones too much. Just take alook at Amstrad, a computer with the same 8 bit Z80a processor as the SVI's still taking the market place.

WE NEED TO GET THESE OFF THE GROUND IF
THIS COMPUTER IS GOING TO SURVIVE.

1. A talking card for the SVI.

2. A MSX-Dos type disk operating system.

3. A Internal Modem for the expander slot.

4. A double sided conversion to the card controlling the disk drives.
   (You need double sided drives for this conversion.)

5. Cheap access to disk drives ($150-$200).

6. Machine Code disassembler for converting MSX games to earlier SVI's.

7. A double sided Basic-Dos operating system.

Any people who are interested please contact me.
SUBJECT TO INTEREST

Mr ROSS BRISBANE.
29 BARRAMAY STREET,
MANLY WEST. Q. 4179.

We thank Ross for sharing the history of computers with us and if anyone else has anything they would like to send in we would be happy to here from you.

SCREEN CONTROL & CURSOR CONTROL
for the
SPECTRAVIDEO SV 806 80 COLUMN DISPLAY
by Noel Green

As supplied, the three SV 80 column systems, CP/M 2.20, 2.22, and SV Extended Disk Basic, leave much to be desired in screen stability etc.

The following exercise is submitted as a suggested "fix" used by the auther to:-

(A) Stabilise the display

(B) Equalise the display for all systems (so you don't have to adjust your monitor)

(C) Choose a cursor form &/or flash speed

I will go in simple steps from the beginning so as to allow novices, like myself, to follow the process, one step at a time, and to get some idea of how to use DDT80 (or DDT).

To make sure that you don't corrupt anything, first take a blank disk, format it, and use Sysgen to place a copy of the CP/M System on it. If you have two disk drives, use Pip (or Sweep) to copy SYSGEN.COM and DDT80.COM (or DDT.COM) onto this "work" disk. If you only have one drive, then probably the easiest way out is to 1COPY your System disk then type ERA [Filename] <ENTER> to get rid of the unwanted files and get some room.

Before you can "hand" modify the System tracs you must first get a copy of it as a file. To do this, you use SYSGEN.COM (on your work disk).

I start with a "clean" Computer. By this I mean TURN THE COMPUTER AND EXPANDER OFF!! then on again. If your System gets corrupted in a minor way, you may never find out but, something will eventually foul up.

Insert the disk you have made in drive A: and type SYSGEN <ENTER>, just as you would to prepare a new disk. When you get the messages:-

```
CP/M Sysgen Version etc.
SOURCE DISK DRIVE? (A or B or ENTER to
skip or Control-C to reboot.)
        Press A.
```

You will now get the message:-

```
SOURCE ON A THEN PRESS ENTER.
```

Press <ENTER> and after the drive has run, you will get:-

```
DESTINATION DISK DRIVE?  (A or B or ENTER to
reboot)
```

At this stage, press ENTER and you will get:-

```
PLACE CP/M DISK IN DRIVE A
AND PRESS ENTER TO REBOOT
```

Press <ENTER>. Immediately you get the CP/M A> or A0> prompt type:-

SAVE 52 CPMSYS.COM and press <ENTER>.

You now have a file on your disk which has an image of Sysgen and your System, CP/M or DISK BASIC on which you can work with DDT80 (or DDT).

Now type:- DDT80 CPMSYS.COM <ENTER> and your terminal will display:-

```
DDT VERS 2.2
NEXT    PC
3500  0100
-
```

Now type D24D0 <ENTER>. (At this time I am using the address for CP/M V2.20. The relevant addresses for the other Systems are tabulated at the end.) The top line of the block displayed should look like this:-

```
24D0 D3 58 FB C9 6D 50 59 0C 1F 02 18 1A 00 07 60 07
                                        .x..mPY.......'.
```

The sequence of values from 24D4 (6D) to 24DF (07) control the output of the SV 806 card. To alter these, type S24D4 <ENTER> and you will get:-

24D4 6D

Type the new value, 6E, at the cursor then <ENTER> and you will get:-

24D5 50

Since this is the No. of columns (80 decimal) and remains unchanged, simply press <ENTER> and you get:-

24D6 59

Continue as above until all required changes have been made, then type a period (.) and press <ENTER>. This will return you to the DDT "-" prompt. [Look after the table at the end of this article for the various values to replace the existing values ...Max.]

SAVING THE MODIFIED SYSTEM

At this stage you have two ways to go. If you type GO or CTRL C, you exit DDT and return to the A> prompt. Now, as before, type SAVE 52 CPM220.COM (FOR V2.20) <ENTER> or another suitable name. Now type CPM220 <ENTER>. You should get the Sysgen sign-on message:-

```
CP/M Sysgen revision xx
SOURCE DISK DRIVE?(A OR B or ENTER to
  skip or Control-C to reboot.)
```

Now, because this Sysgen already has the modified System "attached", press <ENTER>, and the destination drive is requested. Now just follow the prompts and replace the System Tracks of your work disk with the Modified System Tracks. Now switch off/on or reset and the computer will reboot with your Modified System. If you are happy with the result, then CPM220.COM can be used in the future instead of SYSGEN.COM to put the modified system on any disk.

This is my preferred method, and works equally well with Disk Basic. The alternate method is to exit DDT by typing G100 which will immediately give you the Sysgen sign-on, then continue as above. This will put the System on your disk, but you will not have a saved copy of it.

If modifying Steve McNamee's Custom Version, you will need to (A) use his CSYSGEN.COM, and (B) type SAVE 57 CPMXX.COM (or another suitable name).

The following table lists the start of the sequence for the different Systems.

SYSTEM	SEQUENCE START	CURSOR
SV CP/M V2.20	24D4	24DE
SV CP/M V2.22	2591	259B
S McN CUSTOM (ALL)	2130	213A
SV DISK BASIC	2075	207F

(SV DISK BASIC CURSOR BYTE ALSO AT --- 2676)

The sequence I use is:-

6C 50 59 08 26 02 18 1E 00 07 47 07 (the 47 controls the cursor)

However, if you have a monitor that only uses a small part of the screen, you could try:-

6E 50 5B 08 1D 0C 181B 00 09 49 09 and see how it works.

The cursor control is the eleventh number in the sequence, and works as follows:-

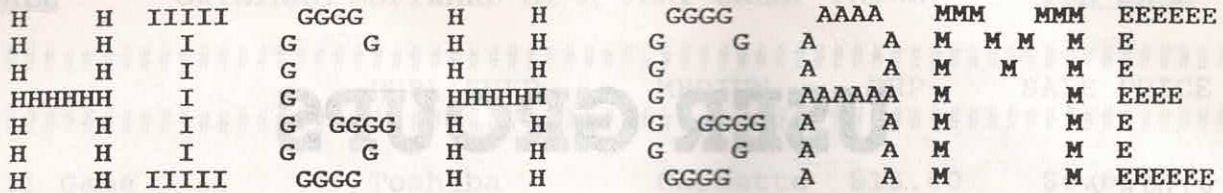
FIRST DIGIT	CURSOR
0	STEADY ON
2	STEADY OFF
4	FAST FLASHING
6	SLOW FLASHING

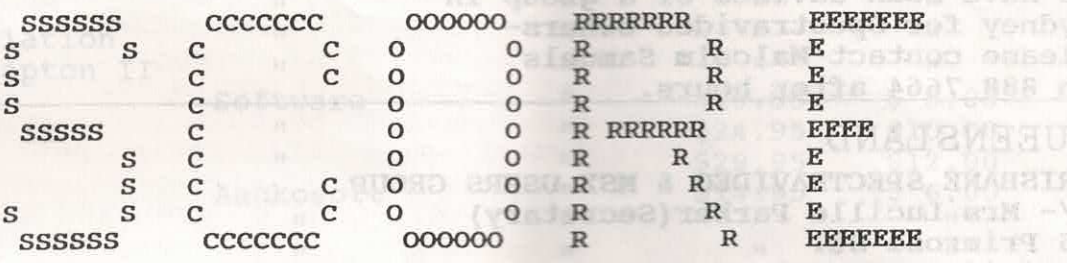
The second digit controls the height of te cursor. eg. 60 gives a full height slow flashing cursor, while I prefer 47 which gives a fast flashing single underline cursor (assuming the previous byte is 07, which is the No. of scan lines per charachter).

SV Disk Basic is a bit more complex, since only some of the System is on the disk, the rest in ROM. Also the keyboard is checked for insert mode after each keystroke and the cursor can be turned off and on (locate,,0/1). The C.R.T. control sequence is at 2075 in the saved System, but a byte at 2676 regenerates the cursor if it has been off, and there may be others. However, all I have been successful at so far is stabilising the display and controlling the on/off\fast/slow part, but not the cursor height. Since my main reason for this exercise was to get rid of the fast flashing full cursor which mesmerised me, I am happy, but if any one knows more, then please let me know.

NOTE: The cursor on the 80 col. screen is **NOT** inverse video as in the 40 col. mode, so it is necessary to have the cursor flashing to see the character behind it, but I find the slow flash more tolerable.

HAPPY COMPUTING... NOEL GREEN





GAME	NAME	SCORE	LEVEL
ALI / FORTY TH	PETER COTTER	80,100	
ATHLETIC LAND	PETER COTTER	324,500	STAGE 25
BATTLESHIP CLAP	JOHN KNIGHT	60,100	
BEAM RIDER	RICHARD JOHNSON	81,112	
BUCK ROGERS	B. RANKIN	122,112	
COMPUT BILLIARD	PETER COTTER	114,300	
DUNGEON ADV-	PHILIP BROUWER	------	LEVEL 3
DORODON	JOHN KNIGHT	45,540	
EXA INNOVA I.E	JOHN KNIGHT	25,710	
GOLF	DAVID CLARKE	EVEN PAR	
GREEN BERET	PETER COTTER	116,300	
H.E.R.O.	JOHN KNIGHT	51,235	
HYPER SPORTS	P. QUINN	475,000	
KING VALLEY	PETER COTTER	49,300	
KNIGHT TIME	CRAIG HURSEY	95%	
LE-MANS	LEE BERGMAN	17,100	
LODE RUNNER	PETER COTTER	279,575	
MANIC MINER	KEVIN TATE	66,663	
MONKEY ACADEMY	JOHN KNIGHT	127,490	
MOUSER	JOHN KNIGHT	35,900	
PIN BALL	B. RANKIN	870,450	
PITFALL 2	P.J. CHRISTIE	199,000	
POLAR STAR	R. DUNCAN	210,000	
RAID BUNGELING	P.J. CHRISTIE	112,030	42 MINUTES
ROAD FIGHTER	CRAIG HURSEY	190,235	STAGE 14
RIVER RAID	D. OLSON	56,650	
SCION	JOHN KNIGHT	131,400	
SENJYO	JOHN KNIGHT	89,150	
SPECTRON	PHILIP BROUWER	9,640	
SUPER COBRA	JOHN KNIGHT	17,690	
SUPER CROSS FOR	JOHN KNIGHT	75,700	
SUPER GOLF	PETER COTTER	-1	UNDER PAR
TRACK & FIELD 1	PETER COTTER	704,630	
ZAXXON	JOHN KNIGHT	52,200	

USER GROUPS

SYDNEY

We have been advised of a group in Sydney for Spectravideo owners- Please contact Malcolm Samuels on 888 7664 after hours.

QUEENSLAND

BRISBANE SPECTRAVIDEO & MSX USERS GROUP
C/- Mrs Lucille Parker(Secretary)
25 Primrose St.
Woodridge,Queensland 4114
---- Phone (07)208 5951 ----

VICTORIA

MELBOURNE MSX & SPECTRAVIDEO USERS GROUP INC.
C/- John Halford
48 Shawlands Avenue
Blackburn South Victoria 3130
---- Phone (03)878 6169 ----

We meet on the first Saturday afternoon of each month (except January) at the Nunawading Library meeting room Whitehorse Rd. Nunawading.

GIPPSLAND MSX USER GROUP
C/- George Francis
31 Donald Street
Morwell Victoria 3840
---- Phone (051)34 5175 ----

WESTERN AUSTRALIA

THE MSX USERS GROUP OF W.A
C/- Max Cowin
PO Box 87 Mt Hawthorn
Western Australia 6016
---- Phone (09)417 3323 ----

Meetings are held on the 4th Tuesday of each month at the Colin st. Hall South Perth at 7.30pm.

NEW ZEALAND

WELLINGTON SPECTRAVIDEO USERS GROUP
PO Box 26-050
Newlands, Wellington
New Zealand 6004

CANADA

SMUG of Montrel
20 - 159th Avenue
N.D. Ile Perrot, Quebec
J7V 5V6 Canada

FINLAND

SVI/MSX Club/Turku
C/- Mika Ristolainen
Lumikonkatu 7H 272 SF-20240
Turka Finland

If any of the above groups wish to enter their meeting times & place or phone number, drop us a line and we will include them in this page.

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Pinball Game	Toshiba	Cassette	$15.00	$6.00
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Battleship Clapton II	"	"	"	"
Jet Set Willy	Software	"	$19.95	$ 8.00
Jet Set Willy II	"	"	$24.95	$10.00
Manic Miner	"	"	$29.95	$12.00
Snake It	Aackosoft	"	$14.95	$ 6.00
Boom!	"	"	"	"
Scentipede	"	"	"	"
Space Rescue	"	"	"	"
Panique	"	"	"	"
Kick It!	"	"	"	"
Chuckie Egg	A 'N' F	"	$19.95	$ 8.00
Turmoil	Bugbyte	"	$12.95	$ 5.00
Ole!	"	"	"	"
Barnstormer	Electric	"	$29.95	$12.00
Chock'n Pop	"	"	"	"
Shark Hunter	"	"	"	"
Sweet Acorn	"	"	"	"
Space Busters	Aackosoft	"	$14.95	$ 6.00
Smack Wacker	"	"	"	"
Type Tutor	Graham	"	"	"
Star Avenger	Kuma	"	$19.95	$ 8.00
Death Valley Goldrush	"	"	"	"
Desolator	Gremlin	"	$34.95	$15.00
Hustler	Bubblesoft	"	$19.95	$ 8.00
Beamrider	Activision	"	"	"
Cyberun	Ultimate	"	$39.95	$16.00
Jack The Nipper	Gremlin	"	"	"
Eggy	Aackosoft	"	"	"
Vera Cruz	Infogrames	"	"	"
Trailblazer	Gremlin	"	"	"
Green Beret	Konami	Cartridge	$49.95	$20.00
Morse Code Tutor	SVI	Cassette	$19.95	$ 8.00
Flight Deck	Aackosoft	"	$44.95	$18.00
Monopoly	Leisure	"	"	"
Scrabble	Leisure	"	"	"

SPECIAL

QUICKNOTES	Sony	Disk	$69.95	$28.00

(In vinyl covered loose-leaf cover with book of instructions-business)

Be quick at these prices they will not last. (Editors comment)

Contact- Gordon A. Browell
13 Brookes street
Biggenden Queensland 4621
Phone : 071-271524

WORDSTORE +

Wordstore plus is a valuable asset to anyone who does any programming at all, it gives your programs the added excitement of speech.

You may have been playing a game and your computer has spoken to you, if so Wordstore + is probabaly the program that produced that speech.

From the time I got the program I was impressed by what it could do, the first thing I did was make a small program up for my daughter's which said a word and they typed it in, if correct the computer would tell them, if not it would ask them to try again, my daughters aged 5 and 7 were very impressed by this machine that spoke to them, and I had to fight them off my computer to get this review out.

I must honestly say that it is a very good program and good value for money.

Wordstore allows the user to record sounds on the tape or with a microphone and convert them to digital data. This data is divided into segments which can be manipulated in different ways including cutting, pasting and reversal.

Sounds are digitally encoded, edited and saved to disk. the data can easily be incorporated into BASIC programs, where it can be easily recalled at different pitches.
The data is only designed to be called up via a user call, so it is stored in part of RAM not used by BASIC, consequently the user RAM remains untouched.

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KNOW WHERE YOU SAW THEIR ADVERTISEMENT.

THE U.K. SCENE

by Brod Brown

Having recently returned from the UK, I thought you might like to hear about my findings. Firstly, I could very easily be persuaded to become a professional tourist. To take in all there is to see in that country would take a lifetime.

I had pre-conceived ideas that the computer market and computer users in England were far more advanced than Australia and that I would be falling over computers and software everywhere I went. Not so. I had to seek it out. This was not easily done, as it was supposed to be a family holiday as well as a business trip. The holiday part consisted of a Country Wanderer package where we hired a car and stayed at country hotels up and down England. That part worked well and we saw some beautiful countryside from Cornwall through to Herriot country in Yorkshire.

The business side of things left a lot to be desired. Businesses in the country did not want to know about computers. In the major centres, if you talked about Spectravideo, people thought you meant Spectrum, and MSX brought blank looks. On the other side of the coin, I saw Commodore and Spectrum software in Woolworths and other large stores, but no hardware. Just as I was giving up, I received a letter from David Webb of MSX Link - the Lincolnshire MSX User Group. He also put me in touch with Bullocks of Birmingham who handle MSX.

Unfortunately, by this time, my schedule did not allow me time to visit either of these people, but I am now catching up on correspondence. You may be interested in what we did have time to discuss.

At last count (December 1986) there were 95,000 MSX users in England. MSX2 is available at selected outlets which are importing them direct. These are selling at £300 ($790) for the basic model. Still no disk drives etc for MSX1. Apparently, computer users in the UK will not look at a basic keyboard unit which sells for more than £200 because they can buy an Amstrad computer/monitor/printer package for £500. The Atari ST (£299) is not selling for this reason.

Some people still have faith in MSX - Bullocks having an MSX expo in November and MSX Link having one in December. Knights of Aberdeen are still heavily into MSX. The loss of MSX Computing magazine has hit MSX users hard, and there is a definite need for COMPUTER FORUM in that country. The big problem is cost.

So, if you are going to UK, take my tip -
Don't combine business with pleasure - both suffer.
Don't expect to be bowled over by computer information.
Don't believe what people tell you about the weather - we were there in Autumn and it only rained for two days while we were there.

A thought just occured to me. If only 10% of those UK users required disk drives which would sell for, say, $300 that would mean a contract worth $3,000,000. If a manufacturer is reading this, we might be in business.

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WITH YOUR GUIDE ALAN CRAWFORD

AFRICA, MARCH, 1903

A dying adventurer staggers (I know the feeling) from the jungle raving about 7 foot tall natives, a searing volcano and a cache of gems fit for a king. ...and so the quest begins.

THE SEARCH FOR KING SOLOMONS MINES

TRADER JIMS OUTPOST, on the ZAMBESI RIVER, JUNE, 1903. We start our adventure with the option of buying four of the six items offered for sale. They are a KNIFE, ROPE, MEDICINES, CANTEEN, MATCHES and RIFLE. Well forget the MEDICINES and the RIFLE has no bullets, so stick with the other four items.

We start out on the bank of the ZAMBESI RIVER overlooking the river and the trading post, you can not get back to the trading post once you have chosen the items offered and started playing. Although you are given a second chance to change your mind, and select four different items before the game proper commences. from here head south to a clearing where you find the body of a dead adventurer. There are two objects here and you will need to drop the knife to be able to pick up both of them. You will need to LOOK twice to find one of the objects and move the body to find the other. From here head back NORTH to where you started then EAST. Be careful when you reach the swamp one more EAST and you are in the quicksand, here you will need to be quick or you will join the other adventurer. To get across the quicksand you should throw the ROPE and then PULL it (a few times) to get to the other side. From here you can not go back.

Once across you will find a GEM and the ROPE, take these with you. From here go SOUTH to the bridge where you find a zulu , I would advise you to KILL him as soon as possible. Then have a good LOOK around and you will see a ledge where you should GO and tie the ROPE to an object that you find there, Its a good idea to leave the gem here, to cut down on the back tracking. From here head back the way you came until you reach the deep ravine where the jungle is. LOOK at the jungle to find the path, thats where you should GO. When you find the undergrowth, CHOP it with the object you found by the dead adventurer, this will open up a GAP which you can go through. EAST of the undergrowth there is a stream, a number of things will need to be done here. The rock can be MOVED with another object you will find if you decide to go swimming or DIVing in the lagoon. The rock will block the stream and the waterfall and open up the cave behind the waterfall. You will also need the VINES, do the same to the vines as you did to the undergrowth to get them. You can also fill your canteen in the stream while you are there.

From the undergrowth GO GAP to the lagoon, if you have blocked the stream you can now GO CAVE. Caution should be taken here, dont be in too much of a hurry. First you will find an amulet, you can READ the amulet but apart from that its not much use. To get past the WEB you must have the matches, destroy the WEB then go south to the inner cave. You only get one move before the web mends itself. Here you will find a box and a cobra, dont worry about the cobra its harmless (trust me). Take the box and return north, dont forget to destroy the web again. Then EAST and NORTH will take you back to the undergrowth, dont forget to pick up the vines and sceptre and any other object that you may have dropped there. Then head back to the bridge.

This is where a lot of people have run into trouble by trying to cross the ravine by way of the bridge. To get across the other side you must go east onto the bridge and the run. You will finish up on the same side but the bridge will now be broken and you will be able to climb the bridge from the far riverbank. GO LEDGE and CLIMB ROPE to get to the riverbank, this is not the riverbank that you want, you will need to cross the river to the far riverbank, but first make sure you have the vines, then cross over the ROCKS to the middle of the river and throw vines they will catch on the other side, then pull vines to get to the other side. Here you will find the dead zulu and the spear that you killed him with, you wont need the spear but you will need the necklace that the zulu is wearing.

The box can be opened by smashing it on the pebbles, (you must be carrying the pebbles) that where on the other bank, inside there is a diary. The information in the diary is not really of any use, and if you do open the box you will not be able to go very much further, so dont worry about opening the box.

Take the box and the necklace and climb the bridge, where you will find a native who blocks your path east. DROP NECKLACE, which the native grabs and then he asks you to bring the serpents nest, which is the box with the cobra wrapped around it,so DROP BOX, the native will then give you an IDOL and then leave, opening up the way east. Before you go any further you will need to be carrying the MATCHES, GEM, IDOL, CANTEEN and SCEPTRE. These are the only items that you will need to take with you and you will need all of them. Then go east into the desert (you can not return once you enter the desert), then go south. You will be getting very thirsty by now, so drink from the canteen and go west to the lake.

You can now discard the canteen. By the lake you will find a palm tree, this should be shaken to release a frond, get the frond and then drop it. It will float on the lake, then GO FROND to get to the other side of the lake. Once there you will see a statue and no visable exits. INSPECTING the IDOL that you should be carrying, will open up a hole in the statue, here you will find some crystals you will need them to cut you bonds when you are captured by the natives and you will be captured by the natives. To continue you should GO OPENING, this will take you to a crevise where you find a hut, an old man and a rope. Give the old man the gem and he will give you a lamp. Get the lamp and the rope and go east where you will be captured by the natives and thrown into a hut while the natives sharpen there knives.

To get out of the hut you should CUT ROPE which you have been tied with and then SMASH HUT to make a hole in the hut, then GO HOLE. You will find yourself by a steep cliff, once here the natives will be hot on your trail. TIE ROPE to SCEPTRE and then throw the rope which catches on the cliff. CLIMB ROPE to get to a windy ledge , dont forget to get the sceptre and rope to stop the natives from following you.

On the ledge there is a small hole, unfortunetly it is not big enough for you to fit through so you will need to make it bigger. This is done by two commands both to do with the SCEPTRE, having worked out what the two commands are, the small hole will now be large enough for you to go through. You will also need to light the lamp before you enter the cave. Inside the cave there is some bats dropping, I didnt worry about getting them, and there is a carving, I could not do much with that and it did not seem to serve any purpose. North of here is the cave with the MYSTIC DOOR. If you try to open the door you will be asked two questions that you must answer correctly. Answer the two questions and then you will be asked if you are brave enough to enter. That is where the game ends, with the comment LOOK FOR PART TWO. This is the worst ending to an adventure that I have ever come across, you dont get the opportunety to enter the door, it just finishes there. The same game on other computers has part two on the same tape as part one. Unfortunetly for M.S.X users we have to buy a second tape to finish the adventure (what a load of bats droppings).

I recieved a letter from TIM JESSEL, VICTORIA. regarding a problem he has with KING SOLOMONS MINES, I'm afraid I was not able to help him out at the moment because the version that he has is SV. and the version on M.S.X are different. The sv. version appears to be the same as the one on commadore so I will make inquiries and when I find out what the main differences are I will publish the sv. version in a later issue. If anybody can help please write in.

I have recieved a couple of letters from readers regarding an arcade game called FUTURE KNIGHT. As I pointed out to those readers, this column is devoted to adventure games and not arcade games. Because adventure games are so time consuming it is not possible to cover more than one field, and adventures are what I prefer. In other words I'm not hot with a joystick. But I am here to try and help with your problems and if I can't I will throw them open to the readers.

So on to FUTURE KNIGHT (arcade type game). I was able to find out a couple of things for readers DALE CLARKE, TASMANIA and P.J.CHRISTIE, VICTORIA. Firstly the CONFUSER should be picked up to gain extra points, but it is not essential that you have it to complete the game. The SAFE PASS is needed to progress through the game and should be kept with you at all times. I was not able to find out anything about the DUNGEONS or how to get past the spaceship level, can anyone be of assistance to P.J. and DALE. In addition I was able to find the cheat program for FUTURE KNIGHT, but I don't know if it works on M.S.X. On title screen press B,U,G,8,7 simultaneously for unlimited lives. I hope that you have some luck with that arcade type game, but in the future stick to the more advanced adventure type game.

P.J. CHRISTIE was also good enough to send in some hints on

SKULL ISLAND. I was going to cover this game very soon, but because P.J. went to the trouble to send in some hints, I have decided to publish his hints this month and conclude the rest of the game at a later stage. So here we go.

To get off the beach, you need to GO WEST from your starting point then 'pace 15' and dig in the sand with a shell the got from underneath the tree, you will find a bottle and if you 'look hole' you will find a rope, which you tie to the anchor by the rowboat. Go to the cliff and throw the rope, It will catch on the ledge. 'climb rope' to the ledge and you will find an axe and some fruit. Go north to return to the beach then go to the tree, chop it three times until it falls then chop again to make rungs in it so you can climb. This brings you out by the lagoon. This will get you started but another tip is dont forget to climb the mast when you reach the shipwreck.

Another problem that someone may be able to help with is, from someone who does not want his name mentioned but his initials are A.C. He wants to know how to turn off the conveyor belts in the game A VIEW TO A KILL, if someone can help me (I mean him) please let me (or him) know.

Well thats about all for this issue, but dont forget to drop me a line if you are having any problems with an adventure and please enclose s.a.e. if you want an answer before the next issue of the magazine. If you have solved a problem that you where stuck on for some time then let me know and I will pass on your solution to other readers. This column is open for any comments that you would like to make about adventures so please drop me a line. Coming soon to these pages will be the conclusion of SKULL ISLAND. I will be trying to find out more about KING SOLOMONS MINES, PART TWO. If anyone out there has any information please let me know. Ive never seen it advertised.

Till next time, bye for now ALAN.

LIST1.PT1

```
990 '====== One Way Bubble ======
1000 GOSUB7010
1010 GOSUB7310
1020 FL=0
1030 FORI=1TONM-1
1040  IFZ[I,1]>Z[I+1,1]
          THEN SWAP Z[I,1],Z[I+1
,1]:                FL=1
1050 NEXTI
1060 IFFL=1THEN1020
1090 N=1:GOSUB7290
1100 GOSUB7220:RETURN
7360 DATA "1..One Way Bubble"
```

LIST2.PT1

```
1990 '====== Two Way Bubble ======
2000 GOSUB7010
2010 GOSUB7310
2020 FL=0
2030 FORI=1TONM-1
2040  IFZ[I,1]>Z[I+1,1]
          THEN SWAP Z[I,1],Z[I+1
,1]:                FL=1
2050 NEXTI
2060 FORI=NMTO2STEP-1
2070  IFZ[I,1]<Z[I-1,1]
          THEN SWAP Z[I,1],Z[I-1
,1]:                FL=1
2080 NEXTI
2090 IFFL=1THEN2020
2100 N=1:GOSUB7290
2110 GOSUB7220:RETURN
7370 DATA "2..Two Way Bubble"
```

SORT - Part 1

by Bruce Goodman.

This is the first in a series of three articles on Sorting. The aim is to describe three different sort techniques each with two variations, and provide a set of programs which will be progressively added so that on completion of the series you will have a program which may be used to compare the merits and disadvantages of the six methods for various applications. While numbers have been used in the example programs, the techniques are equally applicable to strings.

Part 1 discusses the "Bubble Sort" technique, and includes two listings. Bubble sorts require that the list of items to be sorted is already held before the sort starts. The principle of this technique is simply one of 'floating' (hence the name) the smallest item (or largest) to the top (or bottom) of the list, by repetitively comparing each entry in the list with the adjacent entry and swapping their positions if required. The sort is complete when a comparison of each entry with its neighbour is performed and no swaps are necessary.

There are two variatons of this technique, the 'One Way' and 'Bi Directional' Bubble sorts. The first continues to scan the list in one direction floating, say, the smallest items to the top, and the larger ones to the bottom. This will rapidly get the smallest entry to its correct place, but may take many passes to shift the larger items, as these are only shifted one position per pass. The Bi Directional method scans the list in both directions alternatively, thus reducing the total time required.

The first listing, SORT, contains the One Way method at lines 990-1100. The rest of the program provides common support code necessary for all the techniques described in the series - some explanation may be warranted. Lines 10000-10041 set up the key functions and displays, and are written around a disk-based SV 328. Users may need to amend Lines 10011, 10032 and 10036 to suit. Lines 7240-7270 provide a clock function used to time the sorts. Line 160 accepts the number of items you want to sort, while 190-230 set up the screen header.

Lines 7010-7060 place random numbers to be sorted in array Z, and 7440-7470 prints the numbers out in columns. While up to 500 entries can be sorted, only the first 100 will be printed.

The One Way routine sets the swap flag, FL, to zero then scans the array from top to bottom swapping as required. The process is repeated until a complete pass of the array is made and no swaps are necessary, the clock is stopped and the results printed.

The second listing contains the Bi Directional method. (coded as Two Way because Bi Directional is too long for the screen header!) Note that, although the only change from the previous case is the addition of the bottom to top scan, (Lines 2060-2090) significant reductions in the sort times are achieved.

In Part 2, the Insertion methods will be described.

********** SOFT-ADDS **********

THE FOLLOWING PROGRAMS ARE ALL MSX MODE

FTYPE2. (SA1) This is a typing program which may help speed up your typing.

Price $4.00

FIND-ME. (SA2) A game where you play against the computer to find his men before he finds yours.

Price $3.00

LOTTO. (SA3) A game which chooses your numbers and puts them out to the printer or plotter or plays your game. A good program.

Price $3.00

SORT. (SA4) A small utility program that sorts out names or numbers and puts them to a printer or plotter if required.

Price 2.00

<u>These programs are made by ELECTRIC ADVENTURES</u>

PYXIDIS. (SA5) The aliens from the planet Pyxidis are heading towards earth in their huge fleet of war ships. Earth's only hope is you, the most gifted space pilot. But are you good enough to stand up to the alien menace.

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MUNCH MANIA. (SA6) Move Munchie around the maze and try to avoid the ghosts, dont forget to eat the dots and collect the fruit. If you really get into trouble eat a larger dot and then, get those ghosts.
But be carefull your power only lasts a short time.

Price $14.95



MINIMUM CHARGE $12.00 : PLEASE QUOTE (SA) NUMBER WHEN ORDERING.

********** SOFT-ADDS **********

THE FOLLOWING PROGRAMS ARE ALL SVI MODE

D.CUP. (SA7) Although you may never make Wimbledon you may come close with this program.

Price $5.00

HOOKEY. (SA8) This game is based on the game 101 where you must score exactly 101 to win. A game of skill

Price $4.00

GOLF. (SA9) Here is a game of golf for all you people that want the frustration but not the exercise.

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F.PILOT. (SA10) Maneuver your F18 on a mission of seek and destroy. And I hope you have good judgment.

Price $5.00

D.STAR. (SA11) Defend your star base against alien proton blasts A game of skill.

Price $4.00

FORM1. (SA12) Well here you are at the Australian Grand Prix, unfortunately you are in last place. Good luck.

Price $4.00

SUBATT. (SA13) You are the captain of a warship, your mission is to seek and destroy enemy submarines.

Price $5.00

CAVERN. (SA16) A game of endurance where you fly a chopper through caverns and try to rescue trapped men.

Price $7.00

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PYXIDIS. (SA14) The aliens from the planet Pyxidis are heading towards earth in their huge fleet of war ships. Earth's only hope is you, the most gifted space pilot. But are you good enough to stand up to the alien menace.

Price $19.95

MUNCH MANIA. (SA15) Move Munchie around the maze and try to avoid the ghosts, dont forget to eat the dots and collect the fruit. If you really get into trouble eat a larger dot and then, get those ghosts.
But be carefull your power only lasts a short time.

Price $14.95



MINIMUM CHARGE $12.00 : PLEASE QUOTE (SA) NUMBER WHEN ORDERING.

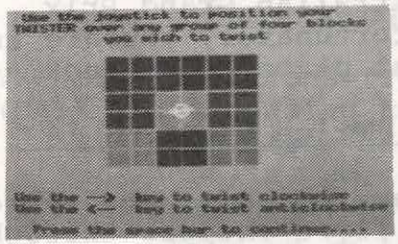
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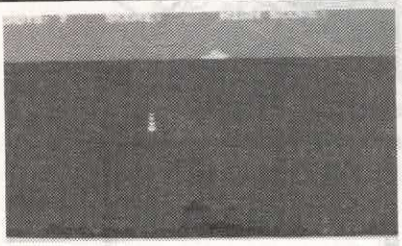
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SVPAK23 - MSXPAK23 -

TWISTER - Like a two dimensional Rubiks cube, this game displays nine different coloured squares each containing four smaller squares. These 36 squares are then jumbled around on the screen and you must rebuild the original display. Its very simple - VERY addictive!

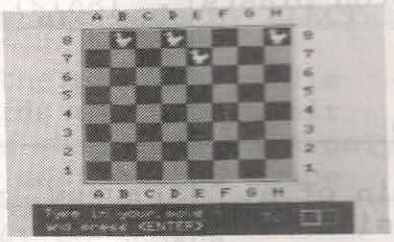
REVERSE - Your task is to arrange the ten jumbled numbers into ascending order by reversing some or all of them. Check it out - its not easy!



TS852 - MSX852 -

GRAND PRIX - You are in the drivers seat in a race to the chequered flag while avoiding slower cars. Hitting another car or two wheels off the track will slow you down and lose points.

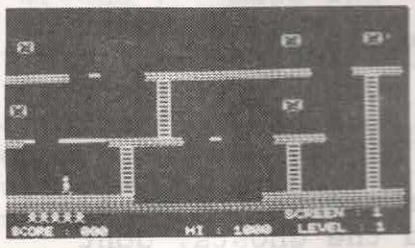
SUBSTRIKE - Drop depth charges on the enemy below. The deeper he is or the faster he moves, the more you will score.



SVPAK24 - MSXPAK24 -

FOX & GEESE - Can you as the Fox penetrate the line of advancing geese and get to the other side of the board before they corner you.

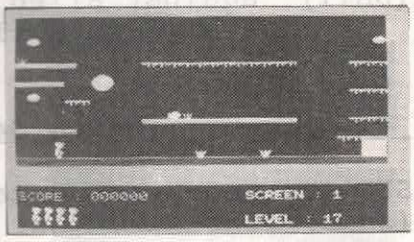
THINK - This is a form of word mastermind. You must guess the word that the computer has selected from the vast number stored in its data banks. This game will certainly make you do as the name suggests.



SVPAK7 - MSXPAK7 -

BOMB SCARE - Very addictive game in which you must climb to different platforms and defuse 5 bombs before your bonus runs out. Has 6 separate screens with the hazards of moving platforms and conveyor belts with holes in them.

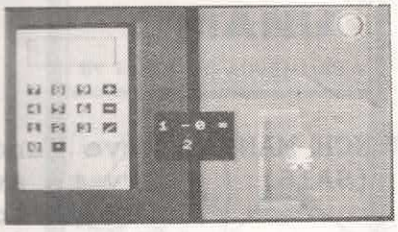
DUNGEON ADVENTURE - A text adventure where you explore the dungeon fighting monsters and gain treasures to discover the location of the great Orb.



SVPAK21 - MSXPAK21 -

GOLDRUSH - Move the miner around the different levels of the mine collecting gold and dodging high explosive charges and falling rocks. The green ledges crumble if you don't get off quickly. When you have collected all of the gold, get to the lift cage for a new screen.

PYRAMID - This is another fascinating text adventure built around the vaults of a pyramid.



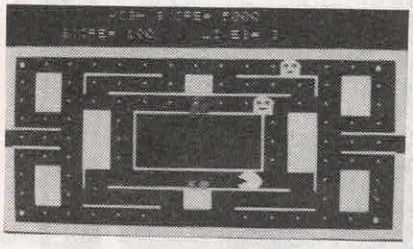
SVPAK25 - MSXPAK25 -

OZMAP - You must drive the car around Australia finding the places required as quickly as you can.

OZQUIZ - Answer general knowledge questions about Australia and her people.

MATH DUEL - Makes practicing maths fun! Has 4 levels of addition and subtraction maths.

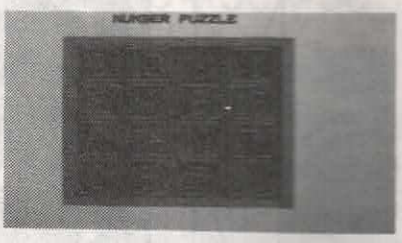
Prices include postage and packing within Australia



SVPAK12 - MSXPAK12 -

MUNCH MAN - Guide your Munchman around the maze eating the dots while avoiding the deadly ghosts. If you eat one of the larger dots you will be able to eat the ghosts for a short while. Gain extra points by eating the fruit that occasionally appears on the screen.

ESCAPE FROM PLANET BETA - This is a text adventure where you are stranded on Planet Beta and must find all the necessary equipment to get off before the planet explodes. Only requires one or two-word commands - eg GET LASER.



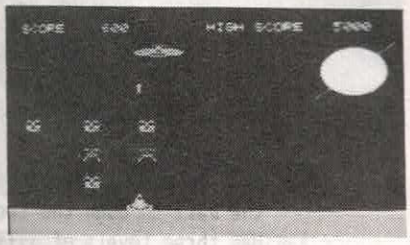
SVPAK20 - MSXPAK20 -

ARTILLERY FIRE - A two player game where each can select a firing angle and velocity to lob shells on his opponent.

NUMBER PUZZLE - Computerised version of the popular hand-held game in which you must slide numbers around into ascending order.

TOWERS OF HANOI - Move all of the discs from peg 1 to peg 3 without placing a large disc on a smaller one.

TENNIS - This is a two player game of Table Tennis.

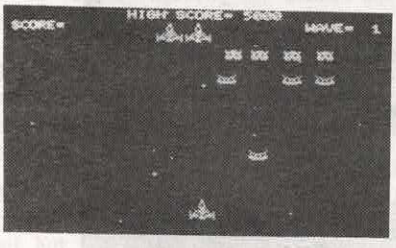


SVPAK9 - MSXPAK9 -

SPACE INVADERS - Shoot the aliens before they invade your planet. Move your ship left and right and shoot missiles at the aliens. Once they have landed the game will end. Score extra points by shooting the UFO which flies overhead.

SUB HUNT - Drop depth charges on the attacking subs and dodge the missiles they fire. You can fire two depth charges at once by firing while moving left or right. Each of the subs is marked with its value.

CHICKEN - Who cares why the chicken crossed the road, we have to find out how! Each time you make it across, the cars speed up.



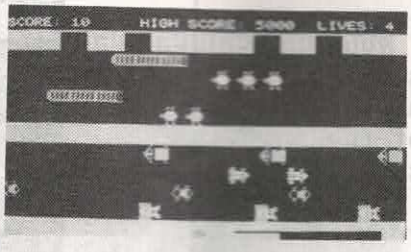
SVPAK5 - MSXPAK5 -

LUNAR LANDER - You are in command of a Lunar Excursion Module with limited fuel but full rotational ability and must land on the lunar surface. Has 8 different landscapes which get harder

GALACTIC ASSAULT - The alien invaders hover above you and one at a time dive on you. Destroy each wave before they get you. For 1 or 2 players.

ROAD RACER - Try and pass as many cars just as fast as you can without crashing.

SPRITE DEFINER - Design sprite shapes on the screen and rotate them when you have finished. Has grids for both sprite sizes.



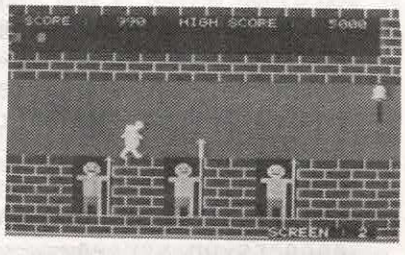
SVPAK6 - MSXPAK6 -

HOPPER - Help Hoppy get across the road (SPLAT!) and then to cross the river (SPLOSH!) and if you are lucky he won't get eaten by the crocs (GULP!!).

SPACE WAR - Challenging game for two players in this duel between your spaceship and that of your opponent. First one to ten points wins.

OTHELLO CHALLENGER - The classic mind game where you must outflank your opponent to capture his pieces. For 1 or 2 players.

PAINTER - Explore the flexibility of your computers colours with this on-screen paint brush.



SVPAK11 - MSXPAK11 -

LUNAR RESCUE - Rescue the stranded astronauts from the moons surface one by one. Dodge the asteroids on the way down and destroy them on the way back up for more points.

SKY BASE - Blast the aliens but watch out for their kamikazi attacks. Shoot two sets and then dock with your mother ship. With this added fire-power go on to destroy the alien base.

QUASIMODO - Can you as the hunchback Quasimodo rescue the beautiful maiden from the French soldiers. Cross the screen avoiding fireballs and spears, then ring the bell to get the next screen.

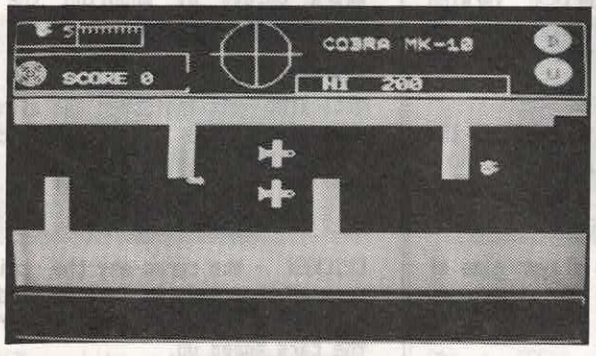
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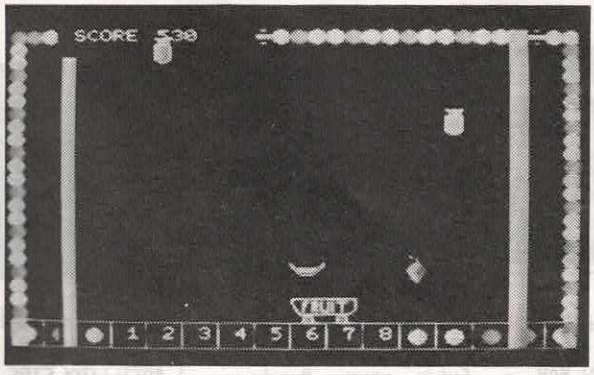
SVPAK26 - MSXPAK26 -

JUMPING COWBOY - A 'platform' game of nine screens where you must collect all the opals to get the next screen.

SUPER SMASHER - Move the bat to smash the ball through the two brick walls. Has three levels of difficulty.

FRUIT CATCHER - Catch falling fruit in your cart. As you catch more, the types of fruit will change and fall at a faster rate.





SVPAK27 - MSXPAK27 -

MIRKWOOD FOREST - Graphics adventure where you must escape from the forest by finding four keys, eating food to maintain energy, while avoiding the spiders. This one will really test your adventure gaming ability.

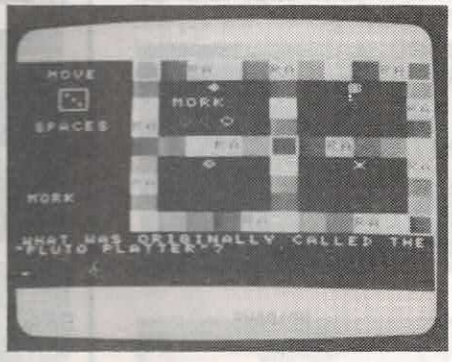
CHOPPER PILOT - The fourteen levels of this game make it very compelling. Keeping clear of canyon walls, fly your helicopter to do battle with enemy fighters, bombers, missiles and UFOs. Great skill required.

UFO (MSX ONLY)

In this machine code game, you are under attack from UFOs. Watch out for the alien unit moving across the bottom of the screen - this will fire 'smart bombs'. Watch out for the meteor shower. Refuel when the truck appears on screen.
Requires 16K RAM, has smooth scrolling, fast response with good sound effects.

METEOR SWARM (SV ONLY)

Another machine code game where your ship is caught in a meteor belt and you must blast your way out. Other ships are caught in the belt and will shoot at you if you do not destroy them first. Comes with bonus game - Birds of Orion.



SPECTRA-TRIVIA (SV 32K)
COMPU-TRIVIA (MSX)

The famous trivia game now on your computer. These are trivia quiz games for up to four players. The computer rolls the dice and each player moves in turn. At certain points you will be asked questions. There is provision for you to create your own question and answer file.

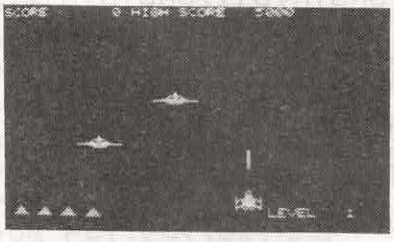
PLEASE SPECIFY SV318/328 OR MSX WITH ORDER.



SVPAK1 - MSXPAK1 -

SPECTRA DERBY - Place your bets and have a day at the races. Even if you lose its painless.

HANGMAN - The classic word guessing game in which you will be gradually hung for each wrong letter.

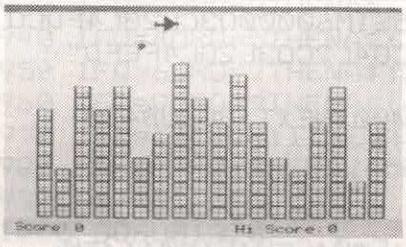


SVPAK10 - MSXPAK10 -

DEMON ATTACK - Shoot the attacking demons as they swoop overhead. You can only destroy a demon when it is fully formed. Killing ten demons gets you on to a harder set.

SEA DIVER - Dodge sea creatures as you make your way to the sea bed. Grab a gold bar and get back to the surface.

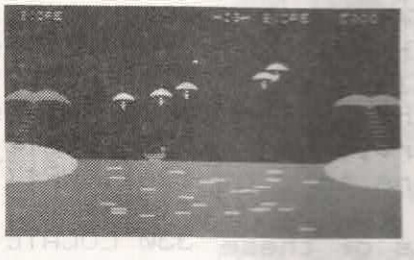
DEATH CITY - Race around the maze trying to shoot your computer opponent but be careful, he is also shooting at you.



SVPAK2 - MSXPAK2 -

SNAKE - Move the snake around to eat the diamonds, but mind you do not cross your own tail.

CITY BOMBER - You must destroy the towering buildings of the city before you can land safely.

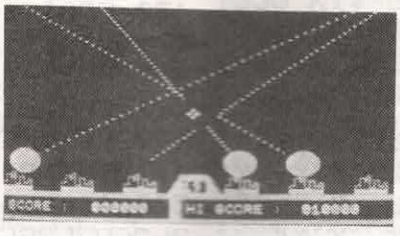


SVPAK8 - MSXPAK8 -

EXPLORER - Watch out for the Pitfalls in this game which will test your control of the joystick. Crocs and other obstacles bar your way as you wander through the jungle in search of treasures.

PARACHUTE - Catch the paratroopers in your boat as they jump from a helicopter into a shark-infested lagoon. The more you catch, the faster they jump.

ANDROID CAVERNS - Move around the caverns of this game and collect the treasure guarded by the androids. But watch out, the androids learn by their mistakes.

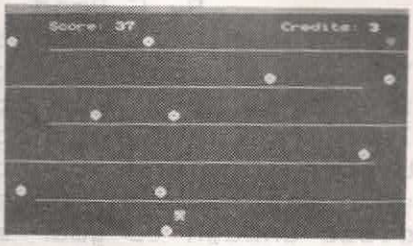


SVPAK22 - MSXPAK22 -

MISSILE COMMAND - Shoot down the incoming warheads before they destroy your cities. You get bonus cities awarded for high scores.

MASTERMIND - Your task is to match the same hidden four-colour combination that the computer has chosen.

LEAKY ROOF - Catch the raindrops leaking through the roof and empty the bucket out of the window before you drown.

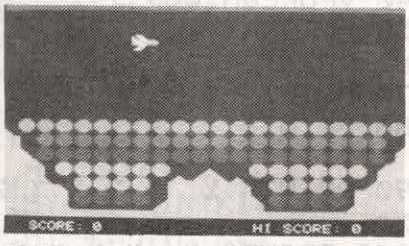


SVPAK4 - MSXPAK4 -

SPECTRA KONG - Win the lady's heart by getting to the top of the screen - but watch out for falling barrels.

TENNIS - A colour version of the original black and white TV game.

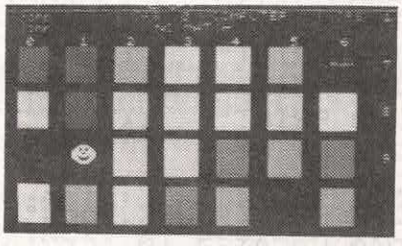
MUSICAL KEYBOARD - This program turns your computer keyboard into a musical keyboard and allows you to select note length and octave.



SVPAK3 - MSXPAK3 -

BOMBS AWAY - Destroy the targets in the ravine below. Be warned - it gets harder the more targets you hit.

BUG CATCHER - Two players move around the garden and the one with the most bugs caught - wins. Great for younger children.



SVEP1 - MSXEP1 -

COPY CAT - The computer will play a set of notes and you must play them back in sequence. There are two levels of play - one note added to the same sequence or a new sequence each time.

MEMORY - A 1 or 2 player game of memory exercise. Uncover two squares on the screen and if the objects revealed match you get another chance

Prices include postage and packing within Australia

```
################
# INVADER ZONE #
################
```

By C.Bond

for M.S.X. Computers

You are trapped in a zone of alien invaders. Move your starfighter left and right to shoot the aliens before they try to get close to your starfighter. Once they are close enough to your starfigther they can use their photons to destroy you.

After shooting two fleets of these aliens you meet up with the alien smother ship.

You must destroy the mother ship to survive the,

INVADER ZONE

```
10 SCREEN1,2:COLOR15,1,1
20 C=2
30 CLS:KEYOFF:GOSUB1550
40 VPOKE&H2004,192
50 PUTSPRITE2,(128,-32),1,2
60 PRINT"%%%                    %"
70 PRINT" %                     %"
80 PRINT" %  %%  % %  %%  %%  %  %
 %"
90 PRINT" %  % % % % % % % % % % %
%"
100 PRINT" %  % % % % % % % % %%%
%"
110 PRINT" %  % % % % % % % % %
%"
120 PRINT"%%% % %  %   %%  %%  %%
%"
130 PRINT
140 PRINT"   %%%%%"
150 PRINT"       %"
160 PRINT"      %   %%  % %   %%"
170 PRINT"     %   %  % %% % %  %"
180 PRINT"    %    %  % %  % %%%%"
190 PRINT"   %     %  % %  % %"
200 PRINT"   %%%%%  %%  %  %  %%"
210 PRINT
220 LOCATE9,16
230 PRINT"By C.Bond"
240 LOCATE7,18
250 PRINT"February 1987"
260 LOCATE,21:PRINT"-Press Space o
r Fire to start"
270 IFSTRIG(0)ORSTRIG(1)ORSTRIG(2)
THEN410
280 TT=TT+1:IFTT<500THEN270
290 CLS:TM=0:LOCATE9:PRINT"SCORING
"
300 LOCATE1,5:PRINTCHR$(&HC2);"  -
  40 Points,-U.F.O.
310 LOCATE1,9:PRINTCHR$(&HC3);"  -
 100 Points,-Star ship
320 LOCATE,13:PRINTCHR$(134);CHR$(
136);CHR$(135):LOCATE1,14:PRINTCHR
$(137)
330 LOCATE4,13:PRINT"- 650 Points,
-Flag ship
340 LOCATE4,17:PRINT"-1000 Points,
-Mother ship"
350 LOCATE,21:PRINT"-Press Space o
r Fire to start"
360 PUTSPRITE2,(20,136),C,2
370 IFSTRIG(0)ORSTRIG(1)ORSTRIG(2)
THEN410
380 C=C+1:IFC=15THENC=2
390 TM=TM+1:IFTM<400THEN360
400 TT=0:CLS:GOTO50
410 PUTSPRITE2,(128,-32),1,2:GOTO4
50
420 BEEP:SOUND0,0:SOUND8,15:SOUND9
,16
430 SOUND12,1:SOUND7,44
440 SOUND11,150:SOUND13,14:RETURN

450 FORI=1280TO1287:VPOKEI,0:NEXT:
VPOKE1280,1
460 FORI=1TO5:A$(I)=CHR$(&HC2):NEX
TI
470 FORI=1TO5:B$(I)=CHR$(&HC3):NEX
T
480 FORI=1TO4:C$(I)=CHR$(133+I):D$
(I)=CHR$(149+I):NEXT
490 CLS:KEYOFF:FORI=1TO80:VPOKE614
4+INT(RND(1)*700),160:NEXT
500 A$="L4T255S1M6000002CL2GL4FDCO
3L2CO2L2GL4FDCO3L2CO2L2GL4FEFD"
510 B$="L4T255S1M6000003CL2GL4FDCO
4L2CO3L2GL4FDCO4L2CO3L2GL4FEFD"
520 PLAYA$,B$
530 FORI=1TO4000:NEXT:GOSUB420
540 GOSUB1320
550 SP$="
                                "
560 MX=1:MY=5:E=0:C=0:EX=0:X3=0
570 X=124:Y=151:D=143:SC=0
580 LOCATE0,0:PRINT"SCORE:"
590 GOSUB2450
600 REM MAIN LOOP
610 PUTSPRITE1,(X,Y),13,1
```

```
620 PUTSPRITE0,(X+1,D),15,0
630 IFINKEY$=CHR$(27)THENSCREEN0:C
OLOR15,4,4:END
640 A=(STICK(0)+STICK(1)+STICK(2))
:IFA=0THENDD=2
650 IFA=3THENX=X+4:DD=0
660 IFA=7THENX=X-4:DD=1
670 IF(STRIG(0)+STRIG(1)+STRIG(2))
=-1THENGOSUB820
680 GOSUB700
690 D=Y-8:GOTO600
700 GOSUB1280
710 CH=VPEEK(6144+(X+3)/8+INT((D)/
8)*32):IFCH=&HC2THEN870ELSEIFCH=&H
C3THEN1020
720 IFCH>133ANDCH<138ORCH>149ANDCH
<154THEN1170
730 IFE=0THENMX=MX+1:GOTO750
740 IFE=1THENMX=MX-1:GOTO800
750 IFMX+11<29THENRETURN
760 IFC=1THENLOCATEMX,MY-4:PRINTSP
$:MY=MY+2ELSE790
770 IFMY=17THENHE=2:GOTO1420
780 E=1:C=0:RETURN
790 C=1:E=1:RETURN
800 IFMX>1THENRETURN
810 E=0:RETURN
820 IFDD=0THENX=X-4ELSEIFDD=1THENX
=X+4ELSEIFDD=3THENX=X+0
830 PUTSPRITE0,(X+1,D),15,0
840 D=D-16:FORI=1TO20:NEXT
850 GOSUB700:IFD>5THEN830
860 RETURN
870 PUTSPRITE0,(0,0),1,0
880 VPOKE6144+(X+3)/8+INT((D)/8)*3
2,168:B=(X+3)/8-MX:Q=INT(B)
890 SOUND13,1:FORG=16TO106STEP5
900 SOUND1,INT(G/10)
910 NEXT
920 SOUND0,0:SOUND1,0:SOUND13,14
930 IFQ=2ORQ=3THENA$(1)=CHR$(32)
940 IFQ=4ORQ=5THENA$(2)=CHR$(32)
950 IFQ=6ORQ=7THENA$(3)=CHR$(32)
960 IFQ=8ORQ=9THENA$(4)=CHR$(32)
970 IFQ=10ORQ=11THENA$(5)=CHR$(32)
980 SC=SC+40:GOSUB2450
990 D=Y-8
1000 GOSUB1320
1010 RETURN600
1020 PUTSPRITE0,(0,0),1,0
1030 VPOKE6144+(X+3)/8+INT((D)/8)*
32,168:B=(X+3)/8-MX:Q=INT(B)
1040 SOUND13,1:FORG=16TO106STEP5
1050 SOUND1,INT(G/10)
1060 NEXT
1070 SOUND0,0:SOUND1,0:SOUND13,14
1080 IFQ=2ORQ=3THENB$(1)=CHR$(32)
1090 IFQ=4ORQ=5THENB$(2)=CHR$(32)
1100 IFQ=6ORQ=7THENB$(3)=CHR$(32)
1110 IFQ=8ORQ=9THENB$(4)=CHR$(32)
1120 IFQ=10ORQ=11THENB$(5)=CHR$(32
)
1130 SC=SC+100:GOSUB2450
1140 D=Y-8
1150 GOSUB1330
1160 RETURN600
1170 PUTSPRITE0,(0,0),1,0
1180 MN=(X+3)/8-MX:O=INT(MN)
1190 BEEP:PLAY"S14M900B"
1200 IFO=3ORO=4ORO=5THENF=F+1:VPOK
E&H2010,240:VPOKE&H2011,240
1210 IFF=3THENGOSUB1440
1220 IFO>8THENFF=FF+1:VPOKE&H2012,
240:VPOKE&H2013,240
1230 IFFF=3THENGOSUB1520
1240 FORI=1TO350:NEXT
1250 GOSUB420
1260 GOSUB1340
1270 D=Y-8:RETURN600
1280 LOCATEMX,MY:PRINTK$
1290 LOCATEMX,MY-2:PRINTK1$
1300 LOCATEMX,MY-4:PRINTK2$
1310 RETURN
1320 K$=SPACE$(1)+A$(1)+SPACE$(1)+
A$(2)+SPACE$(1)+A$(3)+SPACE$(1)+A$
(4)+SPACE$(1)+A$(5)+SPACE$(1)
1330 K1$=SPACE$(1)+B$(1)+SPACE$(1)
+B$(2)+SPACE$(1)+B$(3)+SPACE$(1)+B
$(4)+SPACE$(1)+B$(5)+SPACE$(1)
1340 K2$=SPACE$(1)+C$(1)+C$(3)+C$(
2)+SPACE$(3)+D$(1)+D$(3)+D$(2)+SPA
CE$(21)+C$(4)+SPACE$(5)+D$(4)+SPAC
E$(1)
1350 VPOKE&H2010,128:VPOKE&H2011,6
4:VPOKE&H2012,128:VPOKE&H2013,64
1360 HM=HM+1:IFHM=17THEN1370ELSERE
TURN
1370 FORI=1TO5:A$(I)=CHR$(&HC2):B$
(I)=CHR$(&HC3):NEXT
1380 FORI=1TO4:C$(I)=CHR$(133+I):D
$(I)=CHR$(149+I):NEXT
1390 GOSUB1760
1400 GOSUB1320
1410 RETURN580
1420 BEEP:PLAY"S14M900L64FFF":FORI
=1TO10:COLOR15,6,6:FORN=1TO5:NEXT:
COLOR15,1,1:NEXT
1430 J=X:K=Y:GOTO2480
```

```
1440 F=0:V=0:GOSUB1470
1450 FORI=1TO4:C$(I)=CHR$(32):NEXT
1460 RETURN
1470 BEEP:PLAY"O4T55S14M10G":IFV=0
THENFORI=32TO240STEP16:VPOKE&H2011
,I:VPOKE&H2010,I:GOTO1490
1480 FORI=32TO240STEP16:VPOKE&H201
3,I:VPOKE&H2012,I:GOTO1490
1490 LOCATEMX,MY-4:PRINTK2$:NEXT
1500 SC=SC+650:GOSUB2450
1510 RETURN
1520 FF=0:V=1:GOSUB1470
1530 FORI=1TO4:D$(I)=CHR$(32):NEXT
1540 RETURN
1550 VPOKE&H2010,128:VPOKE&H2011,6
4:VPOKE&H2012,128:VPOKE&H2013,64:V
POKE&H2018,32:VPOKE&H2015,128
1560 SPRITE$(0)=STRING$(8,1)
1570 FORG=1TO4
1580 FORI=1TO32:READQ:A$=A$+CHR$(Q
):NEXT:SPRITE$(G)=A$:A$="":NEXTG
1590 FORI=1552TO1559:READQ:VPOKEI,
Q:NEXT
1600 FORI=1560TO1567:READQ:VPOKEI,
Q:NEXT
1610 FORI=296TO303:READQ:VPOKEI,Q:
NEXT
1620 FORI=1344TO1351:READQ:VPOKEI,
Q:NEXT
1630 FORI=0TO31:READQ:VPOKE1072+I,
Q:VPOKE1200+I,Q:NEXTI:RETURN
1640 DATA1,18,17,16,41,41,43,43,47
,46,44,45,41,16,40,40,192,164,196,
132,202,202,234,234,250,186,154,21
8,74,4,10,10
1650 DATA131,78,50,51,46,77,122,69
,75,81,49,47,65,129,1,1,193,114,76
,204,180,82,190,114,202,134,140,24
4,130,129,128,128
1660 DATA128,0,36,9,6,37,18,7,34,3
,5,40,20,0,65,128,1,0,4,72,0,160,6
4,232,128,112,168,128,40,4,64,1
1670 DATA0,0,0,32,2,2,25,55,67,9,2
0,9,0,40,0,0,0,0,0,0,144,0,160,64,16
0,120,132,4,84,0,0,0,0
1680 DATA36,126,255,153,126,24,36,
195
1690 DATA126,255,165,255,90,153,15
3,66
1700 DATA0,60,78,135,135,135,78,60
1710 DATA0,153,90,0,195,0,90,153
1720 DATA124,66,33,16,8,4,3,1
1730 DATA62,66,132,8,16,32,192,128
1740 DATA60,90,126,255,126,90,90,9
0
1750 DATA219,219,90,90,24,24,24,24
1760 X3=X3+1:IFX3=2THEN1850ELSECLS
:LOCATE12,3:PRINT"READY":GOTO1800
1770 CLS:FORI=1TO90:VPOKE6144+INT(
RND(1)*700),160:NEXT
1780 HM=0:MX=1:MY=11:E=0:C=1
1790 RETURN
1800 A=STICK(1)
1810 IFA=7THENX=X-4
1820 IFA=3THENX=X+4
1830 PUTSPRITE1,(X,Y),13,1
1840 BB=BB+1:IFBB=90THENGOTO1770EL
SE1800
1850 CLS:LOCATE4,10:PRINT"BONUS 10
00 POINTS!!!"
1860 PLAY"L505V15T255DEFGRFEDCDCDC
C"
1870 FORI=1TO10
1880 LOCATE8,5:PRINT"SCORE:5000"
1890 FORU=1TO100:NEXT:LOCATE8,5:PR
INT"           ":FORU=1TO100:NEXT
1900 NEXT:RUN1910
1910 SCREEN2,2:OPEN"GRP:"AS1
1920 PRESET(110,90):PRINT#1,"READY
"
1925 PRESET(5,0):PRINT#1,"SCORE 50
00"
1930 GOSUB1560
1940 TS=170:GS=162:MS=132:RS=128:E
X=0
1950 LINE(110,90)-(200,100),1,BF
1960 ONSPRITEGOSUB2390
1970 FORI=1TO70
1980 X=INT(RND(1)*255)
1990 Y=INT(RND(1)*191)
2000 PSET(X,Y),15
2010 NEXT
2020 DRAW"C11BM30,30D2H2M126,100R4
M225,32E2U1H2L194M30,30"
2030 PAINT(150,30),11:CIRCLE(65,20
),3,6:CIRCLE(80,20),3,6:CIRCLE(95,
20),3,6:CIRCLE(160,21),3,6:CIRCLE(
175,21),3,6:CIRCLE(190,21),3,6
2040 LINE(32,25)-(223,25),6:LINE-(
223,20),6:LINE-(145,10),6
2050 LINE-(138,0),6:LINE-(138,18),
6:LINE-(118,18),6:LINE-(118,0),6:L
INE-(111,10),6:LINE-(32,20),6:LINE
-(32,25),6:PAINT(150,20),6
2060 C=2:PRESET(123,15),1:SPRITEON
2070 PUTSPRITE2,(120,0),C,2
2080 C=C+1:IFC>15THENC=2
2090 A=STICK(1)
2100 IFA=7THENRS=RS-6:MS=MS-6
```

```
2110 IFA=3THENRS=RS+6:MS=MS+6
2120 IFSTRIG(1)=-1THENGOSUB2290
2130 MS=RS+1
2140 PUTSPRITE1,(RS,TS),13,1
2150 PUTSPRITE0,(MS,GS),15,0
2160 GOSUB2180
2170 GOTO2070
2180 PUTSPRITE3,(X,Y),15,0
2190 PUTSPRITE5,(X-15,Y),15,0
2200 Y=Y+20
2210 SOUND0,Y:SOUND1,5:SOUND2,Y:SO
UND3,3
2220 SOUND6,0:SOUND7,(&B00111100)
2230 SOUND8,16:SOUND9,16
2240 SOUND11,10:SOUND12,24
2250 SOUND13,1
2260 IFY>162THENGG=0
2270 IFY>180THENY=30:X=INT(RND(1)*
55)+100
2280 RETURN
2290 PUTSPRITE0,(MS,GS),15,0
2300 PUTSPRITE1,(RS,TS),13,1
2310 A=STICK(1)
2320 IFA=7THENRS=RS-6
2330 IFA=3THENRS=RS+6
2340 GOSUB2180
2350 IFPOINT(MS+5,GS)=11THEN2420
2360 GS=GS-14:IFGS<16THENGG=1
2370 IFGS>0THEN2290
2380 GS=162:RETURN
2390 IFGS<162ANDGS>16THENRETURN
2400 IFGG=0THENHE=1:GOTO2470
2410 IFGG=1THEN2570
2420 LINE(MS+4,GS+15)-(MS+9,GS),1,
BF
2430 GS=162:PUTSPRITE0,(MS,GS),15,
0
2440 RETURN
2450 LOCATE5,0:PRINTSC
2460 RETURN
2470 J=RS:K=TS
2480 PUTSPRITE0,(0,0),1,0
2490 PUTSPRITE3,(255,5),1,0:PUTSPR
ITE5,(255,5),1,0
2500 FORI=0TO13:SOUNDI,I:NEXT
2510 PUTSPRITE1,(J,K),10,3
2520 FORI=1TO80:NEXT
2530 PUTSPRITE1,(J,K),8,4
2540 FORI=1TO80:NEXT:EX=EX+1:IFEX=
6THEN2550ELSE2510
2550 FORI=0TO13:SOUNDI,0:NEXT:BEEP
2560 GOTO2810
2570 FORI=255TO0STEP-10
2580 PUTSPRITE2,(0,0),1,2
2590 SOUND7,254:SOUND8,15
2600 SOUND0,I:SOUND1,2
2610 NEXT
2620 SOUND8,0
2630 FORD=1TO10:NEXT
2640 COLOR15,15,15
2650 FORI=1TO100:NEXT
2660 COLOR15,1,1
2670 SOUND2,5:SOUND3,5
2680 SOUND4,255:SOUND5,15
2690 SOUND6,13:SOUND7,0
2700 SOUND8,16:SOUND9,16:SOUND10,1
6
2710 SOUND11,255:SOUND12,255
2720 SOUND13,0
2730 FORI=1TO60:CIRCLE(128,16),I,1
5
2740 NEXT
2750 FORI=0TO13:SOUNDI,I:NEXT
2760 FORI=0TO13:SOUNDI,0:NEXT
2770 BEEP
2780 SCREEN1,2:LOCATE5,5:PRINT"BON
US 5000 POINTS!!!"
2790 PLAY"L505V15T255DEFGRFEDCDCDC
C"
2800 FORI=1TO2000:NEXT:HE=0
2810 SCREEN1,2:LOCATE9,2:PRINT"GAM
E OVER"
2820 IFHE=0THENLOCATE6,5:PRINT"CON
GRATULATIONS!":LOCATE0,7:PRINT"
 You've survived the":PRINT:PRINT"
        INVADER ZONE":PRINT:PRINT"
    with a perfect score of":PRINT
:PRINT"          10000!!!"
2830 IFHE=1THENLOCATE0,6:PRINT" Mo
ther ship destroyed you.":PRINT:PR
INT"    Your score was 5000"
2840 IFHE=2THENLOCATE0,6:PRINT"  Y
ou were in range of the":PRINT:PRI
NT"     aliens photons.":PRINT:PRI
NT"  Your score was";SC
2850 LOCATE0,18:PRINT"     Another
 game (Y/N)?"
2860 K$=INKEY$:IFK$=""THEN2860
2870 IFK$="Y"THENRUN
2890 IFK$="N"THEN2900
2900 SCREEN0:COLOR15,4,4:KEYON:END
```

MANKALA

A GAME OF STRATEGY

for the SV 318/328

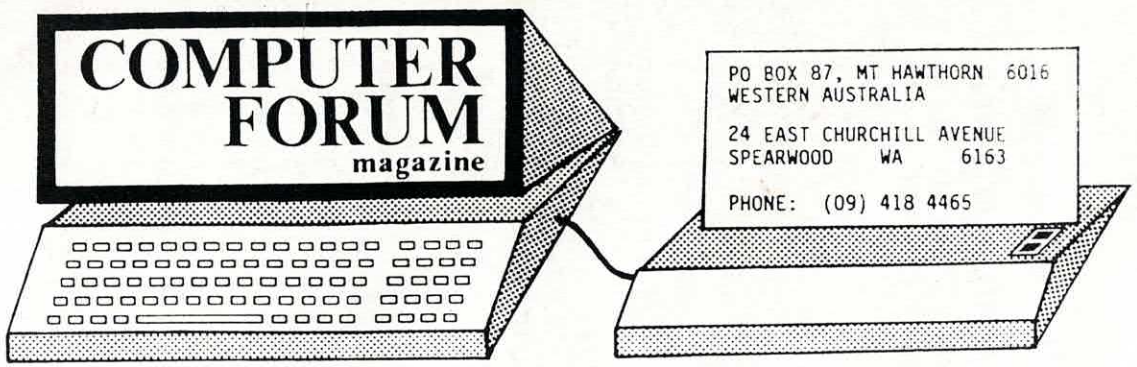
This game is a very ancient one and it originated in Africa. There are two rows of squares and two circles, called 'KALAS', one belonging to each player.

Each square has an equal number of stones. The players take turns to move the stones. As soon as one player manages to clear all five of her/his squares, then that player is the winner.

There is three diferent levels of play for the computer, which can take the place of one player.

```
10 CLS:LOCATE 8,5:PRINT"  MANKALA"
20 PRINT
30 PRINT"a game against the comput
er"
40 PRINT:PRINT"   by D E Rogers
1986"
50 '
60 GOSUB 140 'Initialize
70 ON ST GOSUB 300,490
80 'player's - turn -computer's
90 IF W=0 THEN 70
100 GOTO 820 'Finish off
110 '
120 '
130 '   Initialization
140 DEFINT A-Z ' All variables int
egers
150 DIM A(12)
160 PRINT:PRINT:PRINT"When it is y
our GO, press the number   of the
square which you want to choose."
170 PRINT:PRINT:PRINT"What level o
f play? (1-3)"
180 LEV=VAL(INPUT$(1)):IF LEV<1 OR
 LEV>3 THEN 170
190 PRINT:PRINT"Do you want to go
1st (1) or 2nd (2)?"
200 ST=VAL(INPUT$(1)):IF ST<>1 AND
 ST<>2 THEN 190
210 REM Initialize counters
220 FOR I=1 TO 5
230  A(I)=3: A(I+6)=3
240 NEXT I
250 A(6)=0: A(12)=0
260 GOSUB 1020 'Set up screen
270 RETURN
280 '
290 REM ** player's turn  **
300 WR$="W": GOSUB 1380 'Write GO
310 T=VAL(INPUT$(1)) 'Input turn
320 IF T<1 OR T>5 OR A(T)=0 THEN 3
10
330 WR$="E": GOSUB 1380 'Erase GO
340 GOSUB 1270 'Erase counters
350 GOSUB 920 ' Recalculate counte
rs
360 WR$="W": GOSUB 1270 'print new
 counters
370 FOR I=1 TO 1000:NEXT ' Delay
380 ' check whether any counters l
eft
390 W=1
400 FOR I=1 TO 5
410 IF A(I)>0 THEN W=0
420 NEXT I
430 ' Last counter went in kala?
440 IF I1=6 AND W=0 THEN 300'again
450 ST=2 '=> computer's turn
460 RETURN
470 '
480 '  ** computer's turn **
490 WR$="W": GOSUB 1380 'Write GO
500 FOR I=1 TO 5000:NEXT ' pause
510 ' Check for repeats
520   T=11
530   IF A(T)=12-T THEN 680
540   IF T>7 THEN T=T-1: GOTO 530
550 ON LEV GOTO 650,610,570
560 ' Level 3 - avoid clobbering s
elf
570   T=11
580   IF A(T)>=18-T THEN 680
590   IF T>7 THEN T=T-1: GOTO 580
600 ' Level 2 - clobber other play
er
610   T=11
620   IF A(T)>=12-T THEN 680
630   IF T>7 THEN T=T-1: GOTO 620
640 ' Level 1 - take heap furthest
 from kala
650   T=7
660   IF (A(T)=0) AND (T<12) THEN
T=T+1      : GOTO 660
670 ' Take turn now
680 WR$="E": GOSUB 1270 'erase cou
nters
690 GOSUB 920'Recalculate counters
700 WR$="W": GOSUB 1270: 'print ne
w counters
710 W=2 ' Check for counters left
```

```
720 FOR I= 7 TO 11
730 IF A(I)>0 THEN W=0
740 NEXT I
750 ' Last counter into kala?
760 IF I1=12 AND W=0 THEN 490' loo
p again
770 WR$="E": GOSUB 1380 'Erase GO
780 ST=1 '=> player's turn
790 RETURN
800 '
810 'Conclusion
820 LOCATE 50,40
830 IF W=1 THEN  PRINT: PRINT"CONG
RATULATIONS YOU WON."
840 IF W=2 THEN  PRINT:PRINT "I WO
N THIS TIME."
850 LOCATE 50,120
860 PRINT"DO YOU WANT ANOTHER GAME
?"
870 Q$=INPUT$(1)
880 IFQ$="Y"ORQ$="y"THEN CLS:RUN
890 END 'of main program
900 '
910 ' Recalculate counters
920 N=A(T)' No. of counters to red
istribute
930 A(T)=0
940 FOR I=1 TO N
950 I1=(T+I) MOD 12
960 IF I1=0 THEN I1=12
970   A(I1)=A(I1)+1
980 NEXT I
990 RETURN
1000 '
1010 'Graphics Subroutines
1020 'Set up graphics screen
1030 CLS
1040 SCREEN 1
1050 COLOR 15,1
1060 FOR J=0 TO 1
1070  FOR I=0 TO 4
1080   I1=10+I*50
1090   J1=J*145
1100   LINE (I1,J1)-(I1+45,J1+45),
14,B
1110  NEXT I
1120 NEXT J
1130 CIRCLE(40,95),22
1140 CIRCLE(216,95),22
1150 LOCATE 100,95:PRINT"MANKALA"
1160 LOCATE 20,62:PRINT"Computer's
"
1170 LOCATE 20,125:PRINT"KALA"
1180 LOCATE 190,62:PRINT"Player's"
1190 LOCATE 210,125:PRINT"KALA"
1200 FOR I=1 TO5
1210   LOCATE 50*I-40,147 : PRINT
I
1220 NEXT I
1230 GOSUB 1270
1240 RETURN
1250 '
1260 REM ** print number of counte
rs *
1270 IF WR$="E" THEN COLOR 1 ELSE
COLOR 15
1280 FOR I=0 TO 4
1290   LOCATE I*50+30,165: PRINT A
(I+1)
1300 NEXT I
1310 LOCATE206,90: PRINT A(6)
1320 FOR I=0TO4
1330   LOCATE 230-I*50,20: PRINTA(
I+7)
1340 NEXT I
1350 LOCATE 30,90: PRINT A(12)
1360 RETURN
1370 '
1380 IF WR$="E" THEN COLOR 1 ELSE
COLOR 15
1390 IF ST=1 THEN LOCATE 170,70 EL
SE LOCATE 70,70
1400 PRINT "GO"
1410 COLOR 15
1420 RETURN
```



PO BOX 87, MT HAWTHORN 6016
WESTERN AUSTRALIA

24 EAST CHURCHILL AVENUE
SPEARWOOD WA 6163

PHONE: (09) 418 4465

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